



UPSKILLING PROGRAM: **DATA SCIENCE AND ANALYTICS**

Course: **Database and Business Intelligence with Microsoft Power BI**

Lecturer: **Lay Vathna**

Final Exam - Sales Analysis Report

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I. Introduction

The objective of report is to analyze the sales data to understand the insightful of business performance across different regions, products, and market segments in order to make an informed, data-driven decision-making.

A dataset **"SaleRecord.xlsx"** is used for analysis. It provides detailed insights into sales performance such as sale across regions, market segments, customer, product sold, and profits. The dataset is organized into four main tables: Geography, Product, Sale, and Segment, making it a comprehensive resource for analysis.

II. Data Overview

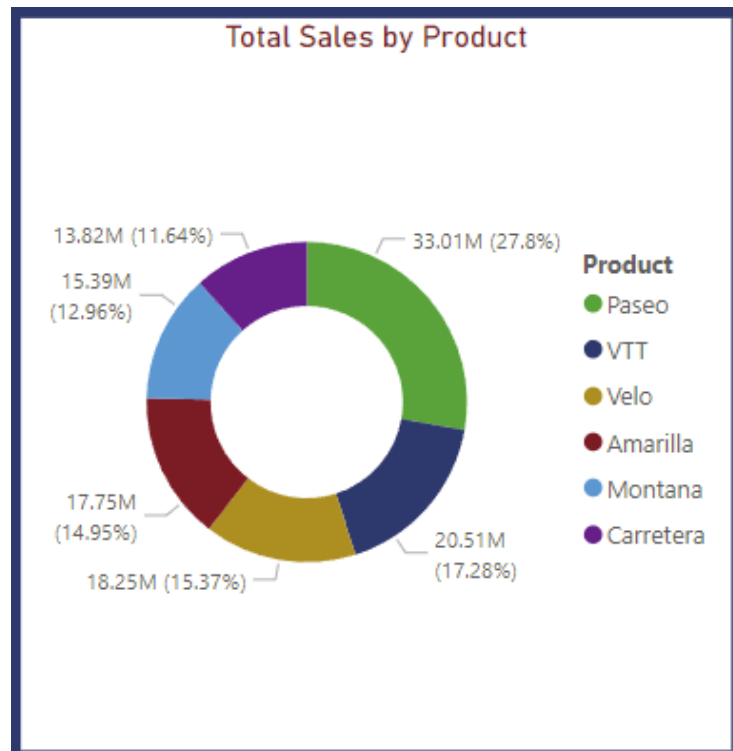
Summary of our sales dataset:

- Total number of sale transactions is 700 records.
- Key metrics:
 - Total Gross Sales: **\$ 127,931,598.50**
 - Total Net Sales: **\$ 118,726,350.26**
 - Total Profit: **\$ 16,893,702.26**
 - Total Transactions: **700**
 - Total Discount: **\$ 9,205,248.24**
 - Profit Margin (%): **14.23 %**
- Software used for data visualization and analysis is **Microsoft Power BI**.

III. Sales Performance Analysis

This section we analyze on sales performance. In 2013 and 2014, **the total sales reached to \$118.73 million**, this includes sales from all the products across the regions. The overall **profit amounted to 16.89 million**, representing the **average 27.90% profit margin**.

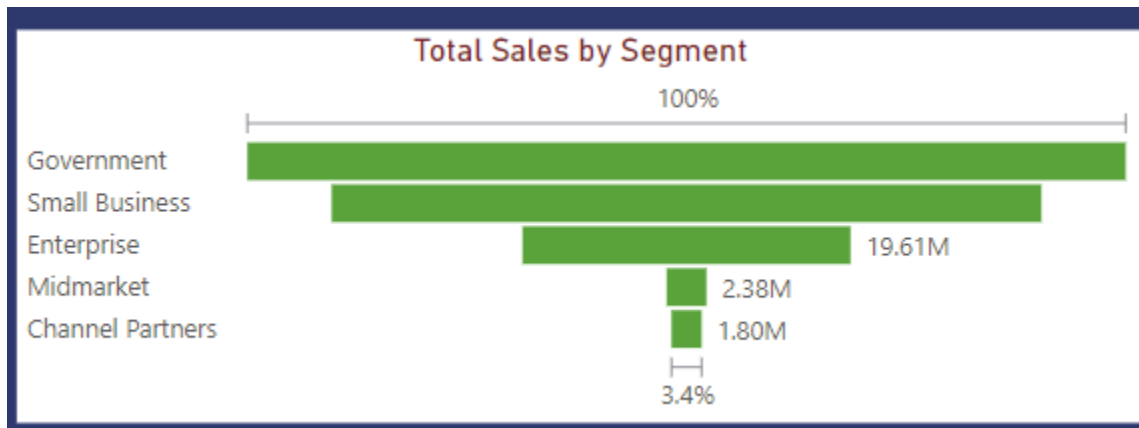
The donut chart illustrated that in **\$118.73 million** of sales are distributed by products: Paseo 27.8%, VTT 17.28%, Velo 15.37%, Amarilla 14.95%, Montana 12.96%, Carretera 11.64%. The product with the most contribute is **Paseo 27.8% equivalent to \$33.01 million**.



The peak Gross sale & Profit is in fourth quarter where the total gross sale amount goes up to \$37.35 million with the total amount of discount \$2.77 million and \$4.96 million of Profit. Compare to the other 3 quarters, it has the most discount which indicates that the peak of gross sale is also related to the product discount.

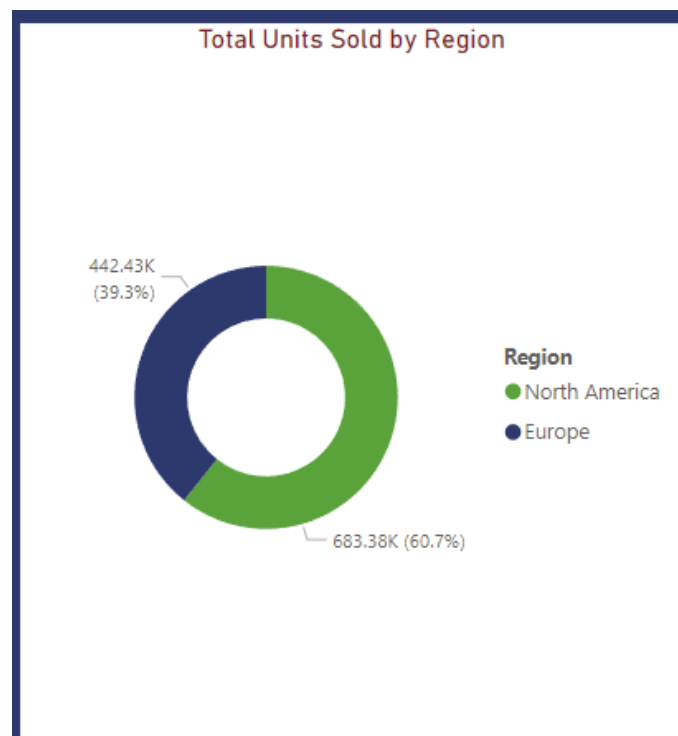


Most of our total sales in ranking from high to low by segments. The highest is Government \$52.50 million then Small Business \$42.42 million, Enterprise \$19.61 million, Midmarket \$2.38 million and lowest is Channel Partners \$1.80 million.

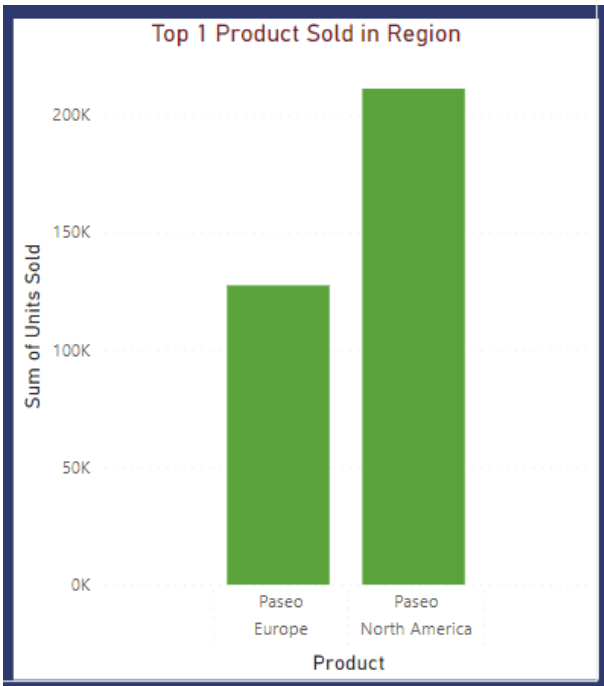


IV. Regional Performance

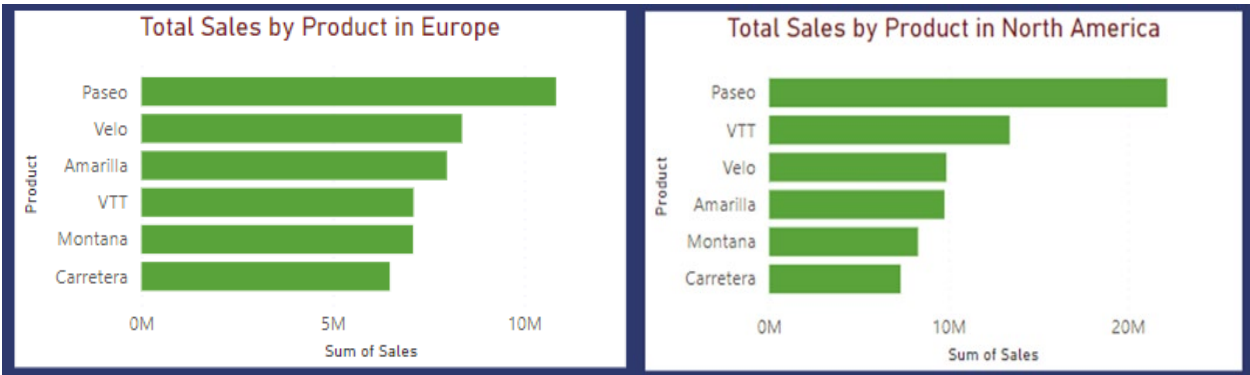
In our sale dataset, there are two regionals, Europe and North America. 442.43K units which equal to 39.3% are sold Europe and 683.38K equal to units 60.7% are sold in North America.



Among the sale units in regions, the top one best seller product in both region is Paseo, 127K units are sold in Europe and 210K units are sold North America. We can conclude that Paseo is the most popular product in Europe and North America.

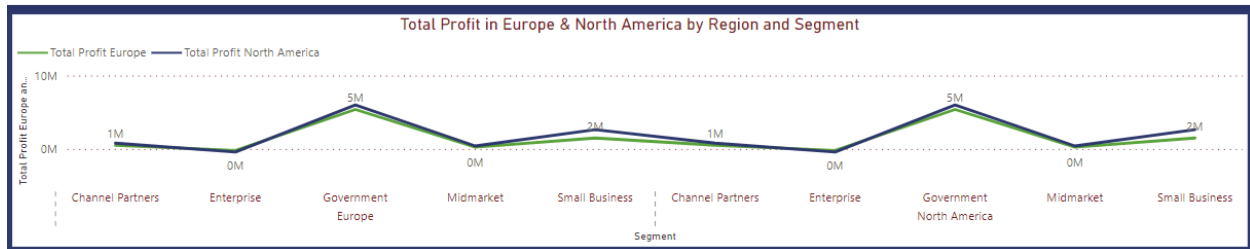


In Europe, the total sale amount rank from Paseo \$10 million, Velo \$8.37 million, Amarilla \$7.97 million, VTT \$7.10 million, Montana \$7.09 million, Carretera \$6.48 million. While in North America, the total sale amount rank from Paseo \$22.18 million, VTT \$13.40 million, Velo \$9.87 million, Amarilla \$9.77 million, Montana \$8.29 million, Carretera \$7.32 million

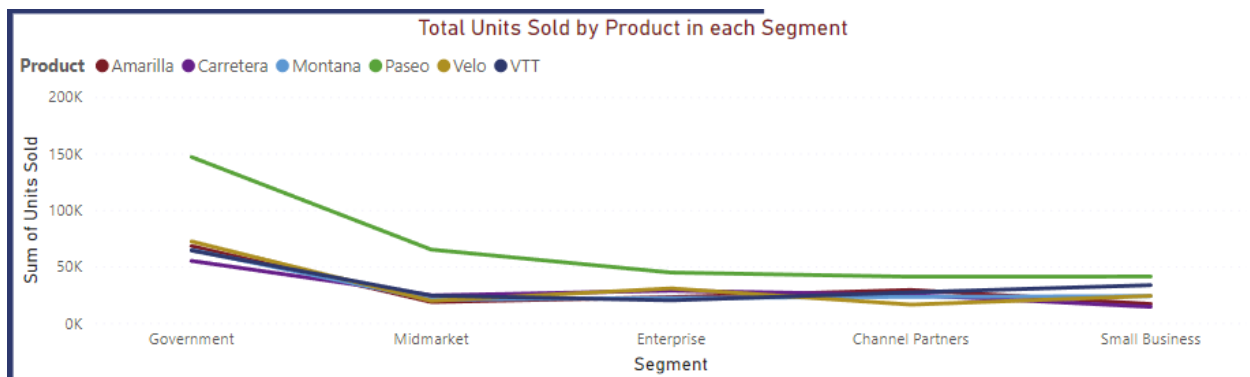


V. Segment Insight

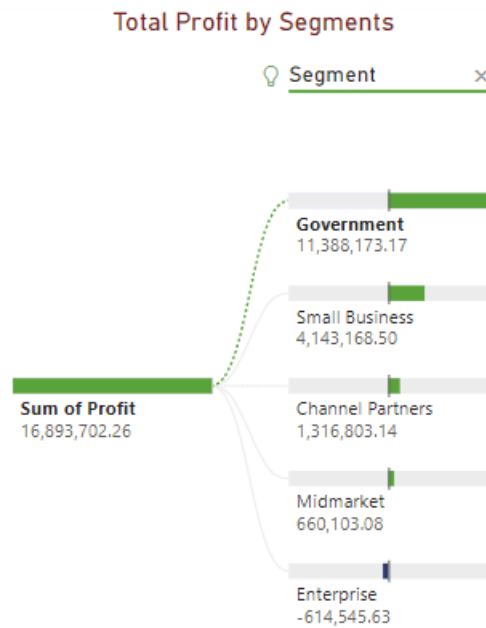
The segment that has the most profit is Government segment for both regions.



Product Paseo has sold 146.79K units under Government Segment.

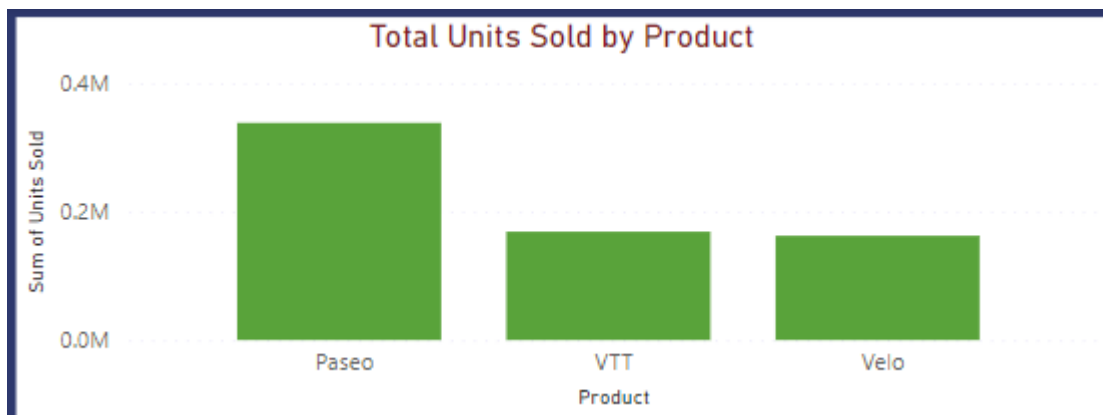


From the composition tree graph below, we see that Government has the most profit \$11.38 million and Enterprise has the least profit \$-614K.

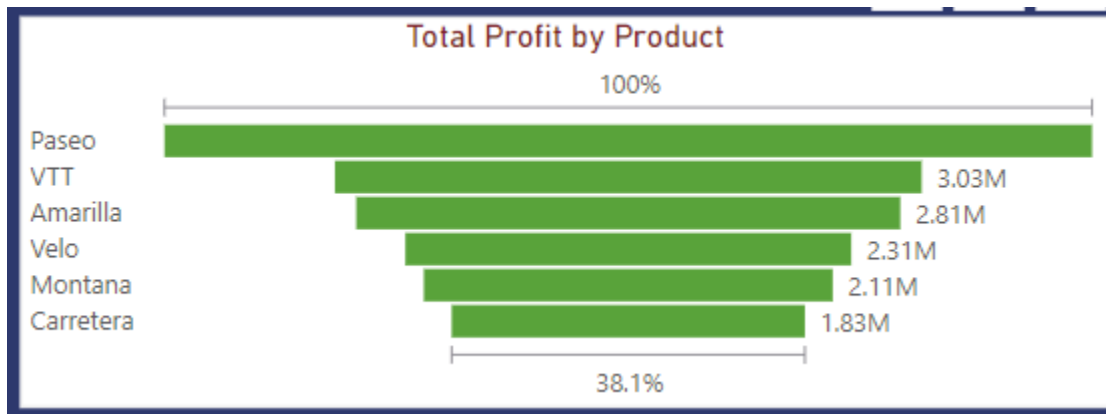


VI. Product Analysis

The top 3 products with the highest units sold are ranking from Paseo, follow by VTT then Velo.



Product that gained the most profit also Paseo with total profit \$4.79 million.

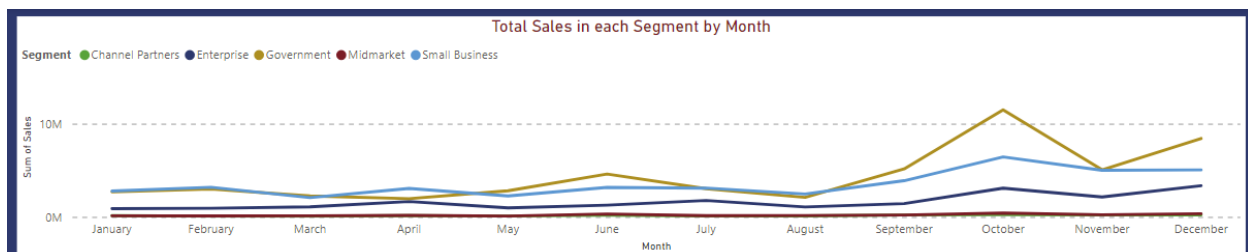


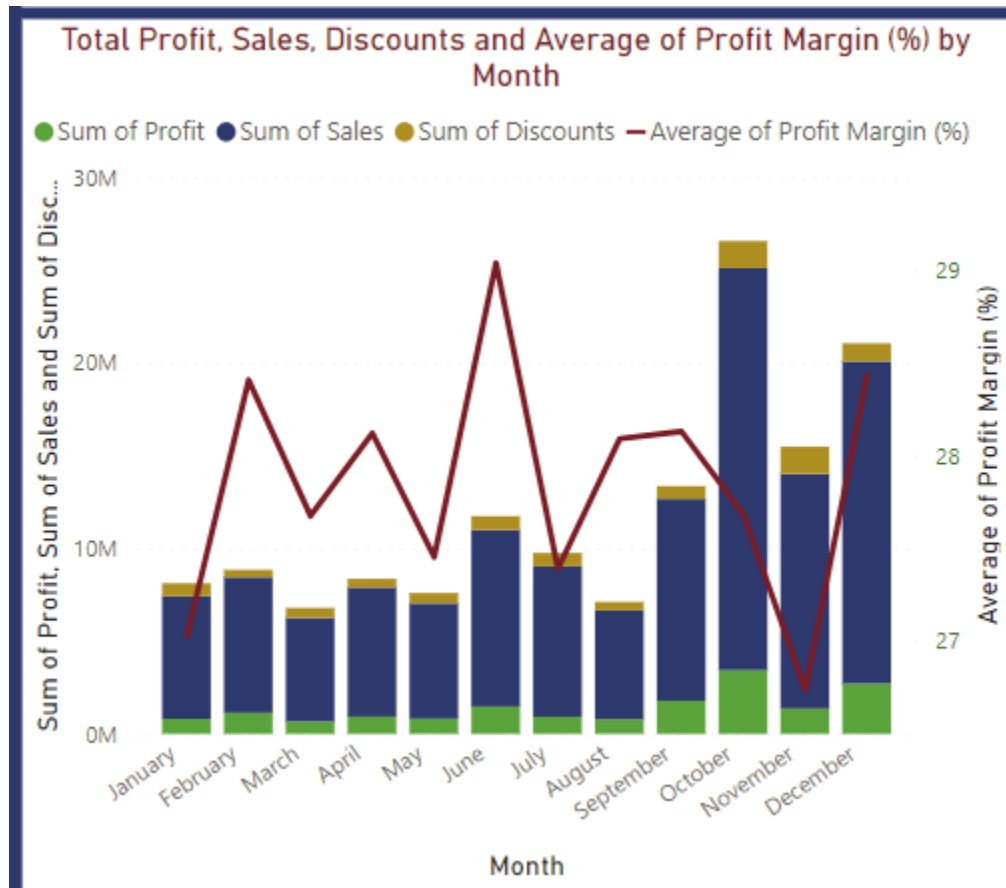
Total Sale by Product

Product	Sum of Sales	Average of Sales
Paseo	33,011,143.95	163,421.50
VTT	20,511,921.02	188,182.76
Velo	18,250,059.47	167,431.74
Amarilla	17,747,116.06	188,799.11
Montana	15,390,801.88	165,492.49
Carretera	13,815,307.89	148,551.70
Total	118,726,350.26	169,609.07

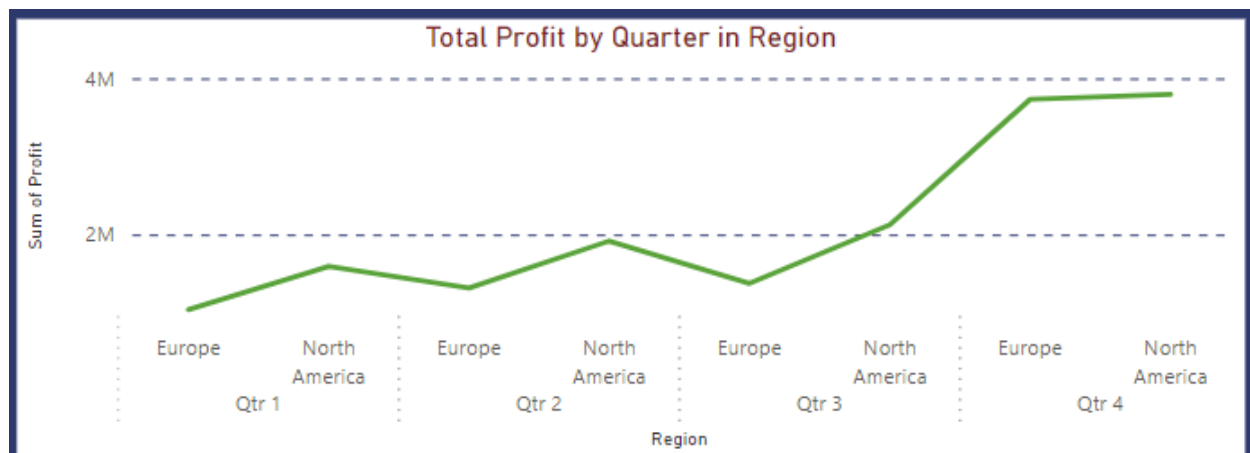
VII. Trends Over Time

October (Quarter 4) is the peak period of sale. We found the most sale is from the Government segment in both Regions which total sale from Government alone equal to \$11.48 million.





In quarter 4, total profit in Europe is \$3.73 million and North America is \$3.79 million.



Among the top 5 customers with the highest sale, 3 are from government segment and one of them with the highest sale like to buy Paseo product.

Top 5 Customers with the Highest Sales					
Customer ID	Region	Segment	Product	Month	Sum of Sales
4154	Europe	Government	Amarilla	October	1,017,338.00
7724	North America	Government	Paseo	July	1,159,200.00
9817	Europe	Government	Velo	October	1,017,338.00
4407	North America	Small Business	Montana	April	1,038,082.50
6802	North America	Small Business	Velo	July	1,035,625.50
Total					5,267,584.00

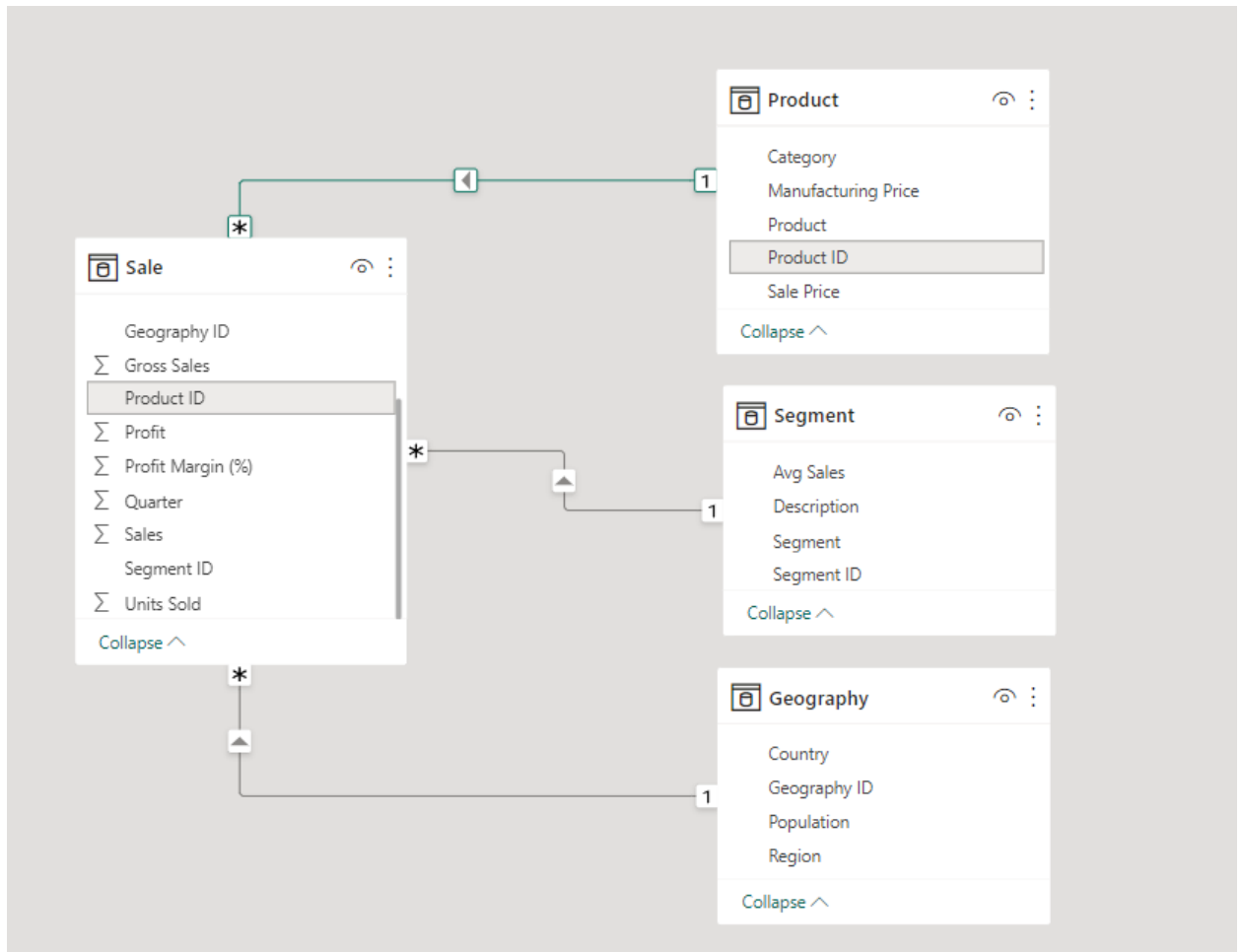
VIII. Actionable Recommendations

Actionable recommendations based on the analysis above are:

- Product Improvement strategies: Focus to keep high performance product Paseo in Government segment. Since sale performance of product Carretera is low, we need to explore on more strategies to increase sell such as apply more discount.
- Profit improvement strategies: optimize the discount on product Paseo as it has highest sale performance.
- Segment Improvement Strategies: Conduct market research in segment Enterprise to understand the market why customers from this segment buy less and understand customer's need.

IX. Appendix

- **Data Modeling**



- **Technical details of the analysis (code snippet)**

DAX Script of Measure "Total Profit Europe"

```
1 Total Profit Europe =  
2 CALCULATE(  
3     SUM(Sale[Profit]),  
4     Geography[Region] = "Europe"  
5 )
```

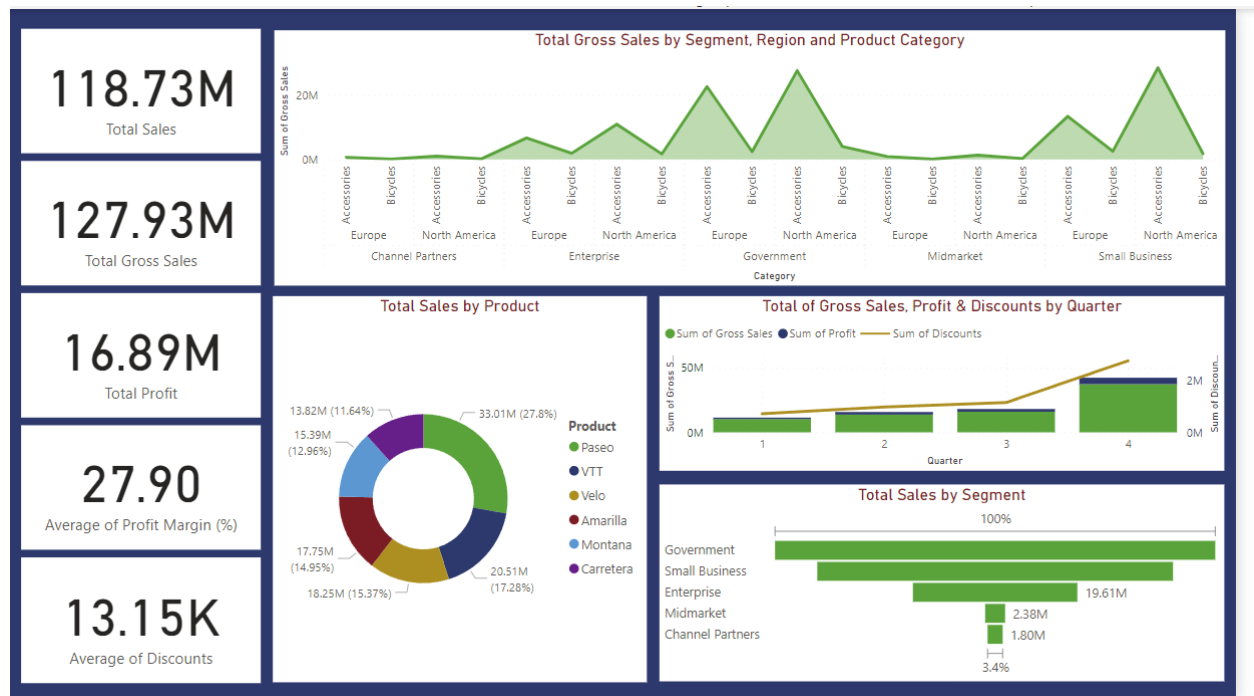
DAX Script of Measure "Total Profit North America"

```

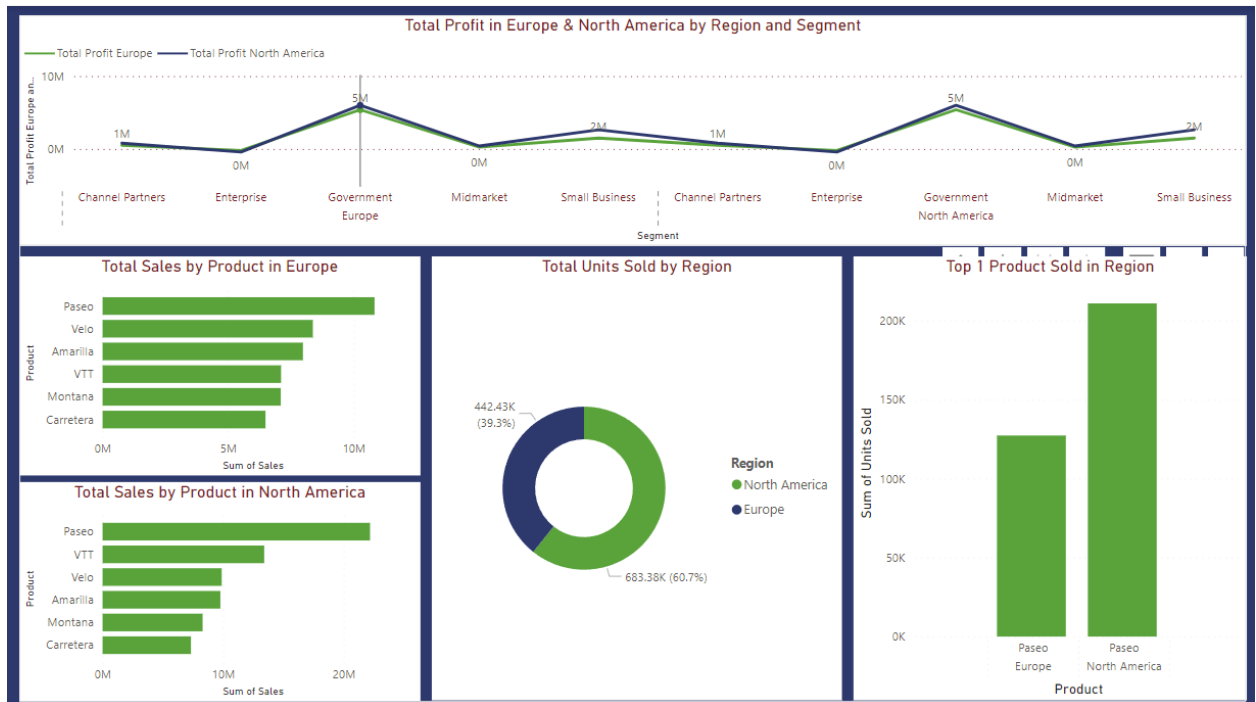
1 Total Profit North America =
2 CALCULATE(
3     SUM(Sale[Profit]),
4     Geography[Region] = "North America"
5 )
    
```

• Detailed Visualization

Page 1: Sales Performance Analysis Dashboard



Page 2: Regional Performance Dashboard



Page 3: Product Analysis Dashboard



Page 4: Trend Over Time Dashboard

